

Transfer Learning for Image Sentiment Classification and Engagement Analysis

Yunxuan Hu & Julia da Silveira Petty

Business Problem Statement

In recent years, the rapid development of digital technologies has fundamentally transformed the interaction between firms and consumers. Digital marketing, broadly defined as the use of online channels and data-driven tools to promote products and build customer relationships, has increasingly shifted toward social media platforms where content is continuously created, shared, and consumed in real time (Krishen et al., 2021; Dolega et al., 2021; Vieira et al., 2022). As users are exposed to increasing levels of information on social media, images offer a cognitive advantage as they can be processed more rapidly than text and are more effective in capturing attention (Zinko et al., 2020). More importantly, images have a stronger capacity to convey emotional signals, as visual cues such as color, composition, and facial expression can directly evoke affective responses and increase engagement (Li & Xie, 2020). Consequently, image-focused social media platforms such as Instagram, Facebook, and TikTok have rapidly expanded, collectively reaching over 5.79 billion active users worldwide, and have become primary channels for firms' digital marketing activities (DataReportal, 2026). Prior research showed that over 90% of brands have official accounts on at least two social media platforms and maintain an active presence (Mitchell et al., 2015). Despite the widespread adoption of social media, firms still face critical challenges in effectively managing and utilizing visual content on social media.

From the perspective of brand-generated content, selecting the most effective images from multiple alternatives represents a critical and challenging decision for firms in fast-paced and competitive social media environments. Prior research shows that brand posting frequency can significantly influence user engagement on social media platforms (De Vries et al., 2012). More specifically, as engagement with social media posts decays rapidly over time, firms that fail to maintain consistent posting quickly lose visibility (Takano & Aida, 2020). Therefore, to maximize engagement within limited time, firms must select visual content that can trigger emotional responses and drive user interaction, thereby enhancing consumer relationships and overall business performance. Prior research suggests that positive emotional appeals, such as happiness and warmth, are associated with higher customer satisfaction and favorable brand evaluations (Bagozzi et al., 1999). At the same time, strategically employing negative emotions, such as fear or anger, can capture user attention and motivate action (Baumeister et al., 2001). However, limited research examined which types of emotional expressions in brand-generated visual content on social media are most effective in driving user engagement. This study aims to help firms address this challenge by providing a systematic approach to identifying the most effective emotional expressions in brand-generated social media content.

From the perspective of user-generated content (UGC), another important challenge faced by firms lies in understanding and utilizing the large volume of image-based content shared across social media platforms. Research demonstrated that UGC plays a critical role in influencing consumer perception, purchase decisions, and brand engagement, as consumers often perceive peer-generated content as more authentic and trustworthy than firm-generated communication (Lou & Yuan, 2019). As social media platforms increasingly encourage users to share visual content, firms are exposed to massive amounts of image-based UGC containing rich emotional and behavioral information. Therefore, effectively analyzing

emotional signals embedded in visual UGC has become increasingly important for understanding consumer attitudes and market responses. However, existing sentiment analysis research has primarily focused on textual data, such as reviews and comments (Pang & Lee, 2008), while visual sentiment embedded in images remains relatively underexplored (You et al., 2015). In practice, firms still lack efficient and scalable tools to automatically analyze and interpret emotions in large-scale image-based UGC. This study aims to help address this challenge by developing a set of professional deep learning solutions to visual sentiment analysis for image-based social media content.

Research Objectives

Building on the challenges identified above, this study aims to develop a systematic deep learning framework for image-based sentiment analysis in social media contexts. Therefore, the first objective is to build a classification model that can automatically categorize visual content into sentiment categories. Specifically, the study considers both binary classification and multi-class classification settings. The binary setting provides an initial benchmark for distinguishing broad positive and negative sentiment, while the multi-class setting further captures more detailed emotional expressions, including happiness, sadness, anger, and melancholy. Due to the complexity of visual sentiment in social media images, the model is expected to achieve moderate accuracy while still providing valuable insights for scalable sentiment analysis and decision-making.

The second objective is to evaluate the effectiveness of different deep learning architectures within real-world settings. In particular, this study compares AlexNet and ResNet-based models under different levels of classification complexity. AlexNet is used as an initial baseline model because of its foundational role in convolutional neural network development (Krizhevsky et al., 2012), while ResNet is introduced as a more advanced architecture capable of learning deeper visual representations through residual connections (He et al., 2016). By comparing model performance across binary and four-class classification tasks, this study examines how model architecture and task complexity jointly affect sentiment prediction accuracy. This comparison also helps identify whether deeper transfer learning models provide meaningful performance improvements in visually diverse sentiment datasets.

The third objective is to connect image sentiment classification with social media engagement analysis. After training the classification models, the study applies the sentiment prediction framework to social media images and examines whether different sentiment categories are associated with different engagement outcomes. Engagement metrics, such as number of likes and number of comments, are used to evaluate whether emotionally distinct images generate different levels of user response. This objective extends the project beyond model development by linking visual sentiment prediction to managerial outcomes that are directly relevant to digital marketing decision-making.

Dataset Overview

This study utilizes two complementary datasets to support model development and real-world evaluation. The first dataset is a publicly available image dataset used for training and benchmarking the visual sentiment classification models under both binary and multi-class settings. The second dataset is a self-collected social media dataset constructed from Instagram brand accounts, which is used to evaluate the performance of the proposed models in real social media marketing context and examine the relationship between predicted image sentiment and user engagement.

Dataset 1: Public Image Sentiment Dataset

The primary dataset used for training the sentiment classification models was constructed based on images collected from the publicly available Unsplash image dataset on Kaggle. Unsplash is one of the world's largest open-source image platforms, containing high-quality photographs collected from diverse real-world environments. Consequently, the dataset covers a broad range of visual contexts, including human portraits, daily-life activities, products, natural landscapes, architecture, and object-centered scenes. In addition, the dataset contains substantial variation in lighting conditions, color composition, background complexity, and contextual information, making it more representative of real-world social media environments.

The original dataset contains 850 images and was organized as a four-class sentiment classification task, including 240 anger images, 238 happiness images, 186 sadness images, and 186 melancholy images. To further compare model performance under different levels of classification complexity, this study additionally constructed a binary sentiment classification setting. Specifically, sadness, anger, and melancholy were merged into a single negative emotion category containing 612 images, while happiness was treated as the positive emotion category with 238 images. This design enables a systematic comparison between binary sentiment classification and more nuanced multi-class emotion classification tasks, allowing the study to evaluate how different deep learning architectures perform under varying levels of emotional granularity and classification difficulty.

Dataset 2: Self-collected Instagram Post from Top 100 Brand Account

To evaluate the real-world applicability of the proposed models, this study further constructed a large-scale social media image dataset based on Instagram brand accounts. Specifically, the dataset was collected from the top 100 brand accounts on Instagram ranked by follower size. To better capture long-term posting behavior and engagement patterns, posts published over a three-year period were collected and analyzed. For each post, the dataset includes visual content information, captions, posting timestamps, number of likes, number of comments, image dimensions, post type, account identifiers, and pinned identifier. These variables provide both visual and behavioral information that supports sentiment prediction and engagement analysis. After data cleaning and preprocessing, the final dataset contains 2,443 social media posts collected from leading brand accounts on Instagram over a three-year period.

This dataset serves two main purposes in the study. First, it provides a realistic testing environment for evaluating the generalizability of the visual sentiment classification models beyond the public training dataset. Second, it enables the analysis of how different predicted emotional expressions relate to user engagement outcomes, including likes, comments, and interaction-related metrics. By combining image sentiment prediction with real-world social media engagement data, the dataset supports the development of practical insights for digital marketing and visual content strategy.

Deep Learning Solutions

To optimize the deep learning solution for image sentiment classification, this study develops a series of deep learning models under different levels of emotional complexity. Study 1 focuses on a relatively simpler binary sentiment classification task that requires the model to differentiate between broad positive and negative emotional patterns. Building on this foundation, Study 2 further increases the classification complexity by requiring the model to distinguish among multiple emotional categories with subtle visual differences. Across both studies, transfer learning frameworks based on AlexNet and ResNet were built, implemented and compared under multiple optimization strategies. Among all models, the AlexNet binary classification model achieved the strongest overall performance, suggesting

that broader emotional distinctions are more suitable for visually diverse social media images. In contrast, although the ResNet-based four-class models produced noticeable improvements over AlexNet in the multi-class setting, distinguishing among visually similar emotional categories remained substantially more challenging. Therefore, the final stage of Study 2 incorporated the Convolutional Block Attention Module into the ResNet framework for further performance improvement.

Study 1: Binary Sentiment Classification

Study 1 focuses on a binary sentiment classification task designed to capture broad positive and negative emotional patterns in social media images. Compared with multi-class emotion recognition, the binary setting provides a relatively simpler classification environment and serves as the foundational stage for subsequent model development.

Data Preprocessing and Augmentation

Before model training, the dataset was divided into training, validation, and test sets using a ratio of 70%, 15%, and 15%, respectively. The training set was used for model learning, the validation set was used for hyperparameter tuning and monitoring model performance during training, and the test set was reserved for final model evaluation.

To ensure compatibility with the pretrained AlexNet architecture, all input images were resized and center-cropped to a resolution of 224×224 pixels. Image normalization was further applied using the standard ImageNet mean and standard deviation values. In addition, several data augmentation techniques were introduced during training to improve model robustness and reduce overfitting. These augmentation strategies included random horizontal flipping, random rotation, and color jittering with adjustments to brightness, contrast, and saturation. By exposing the model to more diverse visual variations during training, the augmentation process helped improve generalization performance on unseen images. For the validation and test sets, only resizing, center cropping, and normalization were applied to ensure consistency during model evaluation without introducing additional randomness.

AlexNet-Based Transfer Learning for Binary Sentiment Classification

Firstly, we adopted a pretrained AlexNet architecture as the foundation of the binary sentiment classification model. The pretrained AlexNet model was initialized with ImageNet weights to leverage visual representations learned from large-scale image recognition tasks. The feature extraction stage of AlexNet consists of five convolutional layers combined with ReLU activation and max-pooling operations. Rather than retraining the entire architecture, the study adopted a transfer learning strategy by fine-tuning only the last convolutional block while preserving the lower-level visual representations learned from the pretrained model. Different learning rates were assigned to different parts of the network, with a lower learning rate of 0.0005 applied to the convolutional layers and a higher learning rate of 0.005 applied to the classifier to stabilize training and improve convergence.

After extracting features from input images, the learned visual representations were flattened and passed into the classifier component of the network. The classifier contains fully connected layers with dropout regularization to reduce overfitting and improve generalization performance. Since the task focuses on binary sentiment classification, the original output layer of AlexNet was replaced with a fully connected layer containing two output neurons corresponding to positive and negative sentiment categories. In addition, because the binary dataset remained imbalanced, class-weighted CrossEntropyLoss was applied during training to reduce the model's tendency to favor the majority class and encourage more balanced prediction performance across sentiment categories.

The model was evaluated with both accuracy and macro F1-score. The final AlexNet binary classification model achieved a test accuracy of 89.71% and a macro F1-score of 0.87, representing the strongest overall performance among all models developed in this study.

ResNet-Based Transfer Learning for Binary Sentiment Classification

To further explore whether deeper architectures could improve classification performance, we implemented a ResNet50-based transfer learning framework. Compared with AlexNet, ResNet50 introduces residual learning through shortcut connections, allowing the network to learn deeper visual representations while alleviating gradient degradation problems in deep neural networks. The pretrained ResNet50 model was also initialized with ImageNet weights and adapted for binary sentiment classification.

The model was first trained with frozen pretrained layers using an initial learning rate of 0.0001 for 10 epochs. After the initial training stage, fine-tuning was performed by unfreezing the upper layers of the network while preserving lower-level visual representations. Specifically, the fine-tuning process started at layer 140, meaning that layers before this point remained frozen while deeper layers were updated during training. A smaller fine-tuning learning rate of 0.00001 was then applied for an additional 10 epochs to stabilize optimization and improve convergence. The original fully connected output layer of ResNet50 was replaced with a binary classifier containing two output neurons corresponding to positive and negative sentiment categories. Weighted CrossEntropyLoss was again applied to reduce the influence of class imbalance and encourage more balanced prediction performance across categories.

The final ResNet50 binary classification model achieved a test accuracy of 85.51% and a macro F1-score of 0.80. Although the ResNet-based framework demonstrated strong feature learning capability and competitive performance, its overall results remained slightly below those of the optimized AlexNet binary classification model. These findings suggest that broader emotional distinctions may be more suitable for comparatively simpler transfer learning architectures in visually diverse social media environments.

Discussion

Overall, the results of Study 1 suggest that pretrained convolutional neural networks are capable of capturing broad emotional patterns from visually diverse social media content. In particular, the AlexNet binary classification model achieved the best overall performance, suggesting that simpler transfer learning architectures may generalize more effectively when the classification task focuses on broader emotional distinctions rather than subtle emotional differences. While binary sentiment classification achieved strong overall performance and effectively reflected broad emotional responses, Study 2 further explores whether deep learning models can distinguish among more nuanced emotional categories with finer visual differences. Both of these approaches could be useful when analyzing the relationship between the sentiment of social media posts, that is why multi-class sentiment classification was also attempted.

Study 2: Multi-class Sentiment Classification

Study 2 extended the task from binary sentiment classification to a four-class sentiment classification setting involving happiness, sadness, anger, and melancholy.

AlexNet-Based Transfer Learning for Multi-class Sentiment Classification

Similar to Study 1, we first implemented an AlexNet-based transfer learning framework for multi-class sentiment classification. The pretrained AlexNet model was initialized with ImageNet weights, and the original output layer was replaced with a fully connected layer containing four output neurons corresponding to the four sentiment categories. We attempted to train the AlexNet model from scratch, but ultimately found that adopting transfer learning

to leverage visual representations learned from large-scale image recognition tasks led to better model performance.

Multiple transfer learning strategies were explored during model development, including feature extraction, unfreezing all fully connected layers, and progressive unfreezing. Among these approaches, fine-tuning the last convolutional block consistently produced the strongest overall performance while maintaining relatively stable training behavior. Different learning rates were also tested throughout the optimization process. Separate learning rates were assigned to the convolutional layers and classifier, with 0.0005 applied to the convolutional layers and 0.005 applied to the classifier achieving the best performance. Lower learning rates and learning-rate schedulers were explored to reduce training oscillation, but none achieved a better performance.

The final model achieved a test accuracy of 63.97% and a macro F1-score of 0.64. Although the model substantially outperformed random guessing, the overall performance remained noticeably lower than that of the binary classification models developed in Study 1. Analysis of the confusion matrix further revealed that happiness was consistently easier to classify, while negative emotions frequently overlapped and were misclassified. These findings suggest that visually similar emotional categories represent a substantially more difficult classification task and motivate the exploration of deeper architectures in the following stage of the study.

ResNet-Based Transfer Learning for Multi-class Sentiment Classification

To further improve performance under the four-class sentiment classification setting, we implemented a ResNet50-based transfer learning framework. The pretrained ResNet50 model was initialized with ImageNet weights and adapted for four-class sentiment classification by replacing the original output layer with a fully connected layer containing four output neurons corresponding to happiness, sadness, anger, and melancholy. Similar to the previous models, transfer learning was adopted instead of training the architecture from scratch. The model was first trained with frozen pretrained layers and subsequently fine-tuned by unfreezing higher-level layers of the network while preserving lower-level visual representations learned from ImageNet. Different learning rates were assigned during training to stabilize optimization and improve convergence performance.

Compared with the AlexNet multi-class model, the final ResNet multi-class model achieved a test accuracy of 68.84% and a macro F1-score of 0.68, representing a noticeable improvement. Nevertheless, despite the improvement, the model still experienced substantial difficulty distinguishing among visually similar negative emotions. These findings motivated the incorporation of an attention mechanism into the ResNet framework to help the model focus more effectively on emotionally informative visual regions and feature channels.

We included the Convolutional Block Attention Module (CBAM) into the final ResNet framework. Specifically, after the pretrained ResNet50 backbone, the extracted feature maps were passed through the CBAM module consisting of both channel attention and spatial attention components. The channel attention module learns which feature channels are more important for sentiment prediction, while the spatial attention module identifies the most informative visual regions within the image. The final attention-enhanced model achieved a test accuracy of 61.59% and a macro F1-score of 0.61, which remained slightly below the performance of the standard ResNet multi-class framework. One possible explanation is that the additional model complexity introduced by the attention mechanism may have increased the risk of overfitting given the relatively limited dataset size and the highly subjective nature of emotion recognition.

Discussion

Overall, the results of Study 2 suggest that multi-class visual sentiment classification is substantially more challenging than binary sentiment classification. Although the ResNet50 framework achieved noticeable improvements over the AlexNet multi-class model, visually similar negative emotions remained difficult to distinguish.

Although the multi-class framework provided more detailed emotional categorization, the overlap among visually similar negative emotions introduced greater uncertainty and potential label noise. In contrast, the binary sentiment framework produced more stable and reliable predictions by focusing on broader emotional distinctions that better reflect overall user emotional reactions toward social media content. Given that the primary goal of the subsequent engagement analysis was to reliably predict image sentiment, we prioritized the binary sentiment framework. Among all binary classification models, the AlexNet framework achieved the strongest overall performance. Therefore, we selected the AlexNet-based binary classification model as our final model for the following engagement analysis.

Image Sentiment and Social Media Engagement Analysis

After completing the sentiment classification stage, the study further examined the relationship between image sentiment and social media engagement. Specifically, the best-performing AlexNet binary classification framework from Study 1 was applied to the Instagram image dataset to classify each post as containing either positive or negative sentiment. The predicted sentiment labels were then incorporated into multiple engagement prediction models together with several control variables, including posting time, post type, account identifiers, pinned status, and comment settings.

To analyze engagement outcomes, separate models were developed for likes and comments. Because the engagement variables exhibited substantial skewness and large numerical variation, logarithmic transformations were applied to both like counts and comment counts to stabilize the distributions and improve model performance. Multiple regression approaches, including Ordinary Least Squares (OLS), Ridge regression, and Lasso regression, were compared to evaluate the robustness of the findings.

The initial regression models achieved relatively strong explanatory performance, with the log-likes model producing an R^2 of 0.44 and the log-comments model producing an R^2 of 0.43. Across all models, positive sentiment was consistently associated with higher engagement outcomes, although the estimated coefficients remained relatively small. At the same time, account-level differences explained a substantial proportion of the variation in engagement performance and were the biggest contributor to predicting engagement.

Video-based content often performs differently in social media settings; therefore, an additional analysis was conducted where this category of posts was omitted to reduce potential noise caused by this video thumbnail data. Therefore, only single image and multiple image posts were kept. In this setting, likes and comments were no longer used as predictive variables for each other due to high correlation, keeping all other control variables the same. The sentiment coefficients remained positive and increased in magnitude; however, the overall predictive performance declined substantially, with the log-likes model achieving an R^2 of 0.01 and the log-comments model achieving an R^2 of 0.03.

Finally, the study reformulated the task as a binary engagement classification problem by categorizing posts into above-median and below-median engagement groups. Logistic regression and Random Forest models were compared using five-fold cross-validation. The Random Forest framework consistently outperformed logistic regression and achieved an accuracy of 66.01% for likes and 63.16% for comments. Feature importance analysis further showed that positive sentiment maintained a consistent positive relationship with engagement outcomes, contributing approximately 3% of the predictive importance in both models.

Therefore, although engagement prediction remained challenging due to the complexity of social media behavior and substantial account-level variation, the findings consistently suggest that positive emotional expressions in social media images are associated with higher engagement performance.

Results & Findings

Overall, the results show clear differences in model effectiveness across architectures and classification tasks. Table 1 summarizes the overall results and key findings across all five models developed in this study.

Table 1. Deep Learning Models Performance Comparison

Study	Base Model	Task	Accuracy	Macro F1	Key Findings
Study 1	AlexNet	Binary	89.71%	0.87	Best overall performance; Broad emotions were highly learnable
	ResNet50	Binary	85.51%	0.80	Strong performance, but slightly below AlexNet
Study 2	AlexNet	Multi-class	63.97%	0.64	Negative emotions frequently overlapped
	ResNet50	Multi-class	68.84%	0.68	Best multi-class performance
	ResNet50 + CBAM Attention	Multi-class	61.59%	0.61	Attention mechanism increased complexity but didn't improve performance

Beyond model development, the engagement analysis further showed a consistent positive relationship between positive image sentiment and social media engagement outcomes. Table 2 showed the results of engagement analysis.

Table 2. Engagement Analysis Results Comparison

Analysis Setting	Model Type	Target Variable	Performance	Sentiment Effect
Full Dataset	OLS / Ridge / Lasso Regression	Log Likes	$R^2 = 0.44$	Positive
	OLS / Ridge / Lasso Regression	Log Comments	$R^2 = 0.43$	Positive
Image and Sidecar Posts	OLS / Ridge / Lasso Regression	Log Likes	$R^2 = 0.01$	Positive coefficient = 0.242
	OLS / Ridge / Lasso Regression	Log Comments	$R^2 = 0.03$	Positive coefficient = 0.196

High vs. Low Engagement Prediction	Random Forest	Above/Below Median Likes	Accuracy = 66.01%	Feature importance $\approx$ 3%
	Random Forest	Above/Below Median Comments	Accuracy = 63.16%	Feature importance $\approx$ 3.2%

Although sentiment was not the only factor influencing engagement, the results consistently suggest that emotional expressions embedded in visual content contribute to user interaction behavior on social media platforms. In particular, positive images in posts were generally associated with higher levels of likes and comments.

Reference

- Bagozzi, R. P., Gopinath, M., & Nyer, P. U. (1999). The role of emotions in marketing. *Journal of the academy of marketing science*, 27(2), 184-206.
- Baumeister, R. F., Bratslavsky, E., Finkenauer, C., & Vohs, K. D. (2001). Bad is stronger than good. *Review of general psychology*, 5(4), 323-370.
- De Vries, L., Gensler, S., & Leeftang, P. S. (2012). Popularity of brand posts on brand fan pages: An investigation of the effects of social media marketing. *Journal of interactive marketing*, 26(2), 83-91.
- Dolega, L., Rowe, F., & Branagan, E. (2021). Going digital? The impact of social media marketing on retail website traffic, orders and sales. *Journal of Retailing and Consumer Services*, 60, 102501.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).
- Krishen, A. S., Dwivedi, Y. K., Bindu, N., & Kumar, K. S. (2021). A broad overview of interactive digital marketing: A bibliometric network analysis. *Journal of Business Research*, 131, 183-195.
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25.
- Li, Y., & Xie, Y. (2020). Is a picture worth a thousand words? An empirical study of image content and social media engagement. *Journal of marketing research*, 57(1), 1-19.
- Lou, C., & Yuan, S. (2019). Influencer marketing: How message value and credibility affect consumer trust of branded content on social media. *Journal of interactive advertising*, 19(1), 58-73.
- Mitchell, A., Gottfried, J., & Matsa, K. E. (2015). Social Media—The Local TV for the Next Generation. *Millennials and Political News*.
- Pang, B., & Lee, L. (2008). *Opinion mining and sentiment analysis*. Now Publishers Inc.
- Takano, C., & Aida, M. (2020). Decay characteristics of user dynamics in online social networks. *IEEE Access*, 8, 73986-73991.
- Vieira, V. A., Severo de Almeida, M. I., Gabler, C. B., Limongi, R., Costa, M., & Pires da Costa, M. (2022). Optimising digital marketing and social media strategy: from push to pull to performance. *Journal of Marketing Management*, 38(7-8), 709-739.
- You, Q., Luo, J., Jin, H., & Yang, J. (2015, February). Robust image sentiment analysis using progressively trained and domain transferred deep networks. In *Proceedings of the AAAI conference on Artificial Intelligence* (Vol. 29, No. 1).
- Zinko, R., Stolk, P., Furner, Z., & Almond, B. (2020). A picture is worth a thousand words: how images influence information quality and information load in online reviews. *Electronic Markets*, 30(4), 775-789.

Appendix

[Study 1: AlexNet-Based Transfer Learning for Binary Sentiment Classification](#)

[Study 1: ResNet-Based Transfer Learning for Binary Sentiment Classification](#)

[Study 2: AlexNet-Based Transfer Learning for Multi-class Sentiment Classification](#)

[Study 2: ResNet-Based Transfer Learning for Multi-class Sentiment Classification](#)

Study 1: AlexNet-Based Transfer Learning for Binary Sentiment Classification

Table 1. Detailed Results of the AlexNet Binary Classification Model

Emotion Category	Precision	Recall	F1-score	Support
Negative	0.92	0.94	0.93	100
Positive	0.82	0.78	0.80	36
Macro Avg	0.87	0.86	0.87	136
Weighted Avg	0.90	0.90	0.90	136

Table 2. Confusion Matrix

	Predicted Negative	Predicted Positive
Actual Negative	94	6
Actual Positive	8	28

Model Training Code in Google Colab

https://colab.research.google.com/drive/1t6zKVfxvfhz_xSAfTK9yoQlkFgZWSP9l?usp=sharing

Study 1: ResNet-Based Transfer Learning for Binary Sentiment Classification

Table 3. Detailed Results of the ResNet50 Binary Classification Model

Emotion Category	Precision	Recall	F1-score	Support
Negative	0.86	0.95	0.91	101
Positive	0.81	0.59	0.69	37
Macro Avg	0.84	0.77	0.80	138
Weighted Avg	0.85	0.86	0.85	138

Table 4. Confusion Matrix

	Predicted Negative	Predicted Positive
Actual Negative	96	5
Actual Positive	15	22

Model Training Code in Google Colab

https://colab.research.google.com/drive/1238aD57duQd5uUSKGHq_Uv7Ku6vrsLb?usp=sharing

Study 2: AlexNet-Based Transfer Learning for Multi-class Sentiment Classification

Table 5. Detailed Results of the AlexNet Multi-class Classification Model

Emotion Category	Precision	Recall	F1-score	Support
Happy	0.69	0.69	0.69	36
Angry	0.60	0.58	0.59	36
Melancholic	0.62	0.75	0.68	28
Sad	0.65	0.56	0.60	36
Macro Avg	0.64	0.65	0.64	136
Weighted Avg	0.64	0.64	0.64	136

Table 6. Confusion Matrix

	Predicted Happy	Predicted Angry	Predicted Melancholic	Predicted Sad
Actual Happy	25	7	2	2
Actual Angry	5	21	4	6
Actual Melancholic	2	2	21	3
Actual Sad	4	5	7	20

Model Training Code in Google Colab

https://colab.research.google.com/drive/1v42U7ZFN5_b9alndT0zb5liZZbvJfLbg?usp=sharing

Study 2: ResNet-Based Transfer Learning for Multi-class Sentiment Classification

Table 7. Detailed Results of the ResNet Multi-class Classification Model

Emotion Category	Precision	Recall	F1-score	Support
Happy	0.76	0.86	0.81	37
Angry	0.65	0.61	0.63	36
Melancholic	0.74	0.59	0.65	29
Sad	0.62	0.67	0.64	36
Macro Avg	0.69	0.68	0.68	136
Weighted Avg	0.69	0.69	0.69	136

Table 8. Confusion Matrix

	Predicted Happy	Predicted Angry	Predicted Melancholic	Predicted Sad
Actual Happy	32	5	0	0
Actual Angry	6	22	1	7
Actual Melancholic	0	4	17	8
Actual Sad	4	3	5	24

Model Training Code in Google Colab

<https://colab.research.google.com/drive/1mOXXkqLKUMBVzYk6tj2OA5-cuWL50G9u?usp=sharing>